ACE ENGINEERING COLLEGE (AN AUTONOMOUS INSTITUTION)

Department of Computer Science & Engineering (IoT & AI&DS)

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MAJOR PROJECT REVIEW — PHASE 1

Project Title: WORKUP: Virtual Professional Experience Platform

A Simulated Workplace Platform for Practical Software Engineering and Skill Verification

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Date: 17-09-2026 | Department: AI&DS

PROBLEM IDENTIFICATION

Describe the Identified Problem

Major challenges students face when moving from academic coding to corporate engineering.

CHALLENGE 1

Clean-Slate Learning (Toy Problems)

Students usually practice writing small code snippets from an empty file. In real software jobs, developers almost never write code from scratch. Instead, they must read, understand, debug, and improve large existing project codebases with existing technical debt.

CHALLENGE 2

No Real Architectural Context

Classroom assignments only give algorithmic prompts. They lack real business background, task tickets, and Architecture Decision Records (ADRs). As a result, students do not learn how to follow company design standards or handle production constraints.

CHALLENGE 3

Lack of In-Depth Code Review

Standard learning websites only use simple pass or fail checks on test cases. Students rarely experience line-by-line code reviews from senior engineers to learn clean coding habits, edge-case handling, and proper pull request communication.

CHALLENGE 4

Unverifiable Resume Claims

Resumes only show self-written lists of technologies and course names. Recruiters and faculty members have no practical way to verify a student's actual problem-solving and coding ability on realistic engineering tasks.

Abstract

A virtual workplace platform bridging the gap between classroom learning and industry practice.

Most computer science students learn programming concepts through lectures and isolated coding exercises, but they often face difficulties when transitioning to professional software companies. In real engineering teams, developers rarely write software from scratch. Instead, they work on pre-existing codebases, resolve project task tickets, follow architectural guidelines, write automated tests, and submit pull requests for code review. Existing learning platforms focus heavily on syntax and algorithmic puzzles, leaving students with limited exposure to realistic team workflows.

This project presents WORKUP, a web-based virtual professional experience platform designed to bridge this practical gap. Instead of solving toy problems, learners enter simulated tech companies (such as FinFlow Technologies) and take on realistic engineering roles. The platform organizes work into project backlogs using a task dependency engine, linking each task to real business scenarios and Architecture Decision Records (ADRs). Learners work inside isolated browser-based development sandboxes where they can edit code, run automated tests locally, and submit their changes for review.

An automated AI-assisted code review engine evaluates code changes against defined rubrics and provides line-level feedback to guide the learner. As tasks are approved, the platform creates an Evidence-of-Skill portfolio backed by verified Git commits. The system is designed and prototyped using Next.js 14, React.js, TypeScript, Prisma ORM, and Vitest, offering a safe, hands-on environment where students gain realistic workplace experience before starting their careers.

Literature Survey

Careful review of published research in software engineering simulation, graduate skills, and cloud sandboxes.

Title / year	Main Findings:	Remarks
Title: Comprehensive Evaluation of an Educational Software Engineering Simulation Environment **Year: 2007 (IEEE TSE)** **Authors: E. Navarro & A. van der Hoek**	• Active simulation helps students understand software engineering trade-offs much better than lectures alone. • Simulated roles and sprint backlogs allow students to see how their decisions affect code quality and project progress.	• Confirms that simulated workplaces are effective for teaching practical software workflows. • WORKUP adopts this concept to simulate realistic company environments and task backlogs.
Title: Gaps Between Industry Expectations and University Preparation in Software Engineering **Year: 2013 (ACM SIGCSE)** **Authors: A. Radermacher & G. Walia**	• Industry survey across 50+ tech companies identified that graduates struggle most with legacy code, testing, and Git reviews. • Universities focus heavily on individual clean-slate coding rather than team-based code maintenance.	• Directly supports WORKUP's focus on legacy codebase tasks rather than simple clean-slate coding. • Shows the need for task tickets with automated testing and pull request reviews.
Title: Lightweight Container Isolation for Computing Education **Year: 2018 (IEEE Trans. Education)** **Authors: N. Kratzke**	• Browser-accessible isolated sandboxes remove local setup friction and prevent student code from escaping to the host server. • Ephemeral execution sandboxes provide a cost-effective way to host programming labs.	• Informs WORKUP's sandbox design, using path-jailed execution environments with strict execution timeouts. • Ensures safe and isolated in-browser code execution for learners.

End Users / Target Beneficiaries

Key roles and user groups that interact with the WORKUP platform.

STUDENTS / LEARNERS

Engineering Students & Career Starters

Explore professional career paths (Software Engineering, DevOps, AI/ML), enter simulated tech companies, work on realistic task tickets, write code in cloud sandboxes, run automated tests locally, receive line-level feedback, and build a verified portfolio of real code commits.

EDUCATORS / CONTENT AUTHORS

Faculty & Curriculum Designers

Create simulated companies, design multi-step project task backlogs (DAGs), set up legacy codebases, define automated test suites, and configure evaluation rubrics to assess student learning with zero manual grading overhead.

RECRUITERS / INDUSTRY MENTORS

Hiring Managers & Senior Engineers

Review candidate portfolios backed by verified Git commits and test reports, evaluate practical problem-solving ability, and provide mentor feedback without relying on self-reported resume bullet points.

Existing System

Overview of current learning platforms, coding websites, and job simulations.

- **Algorithmic Practice Websites:** Platforms like LeetCode and HackerRank focus mainly on isolated algorithmic puzzles and syntax questions, without exposing learners to multi-file projects.
- **Video-Based Online Courses:** MOOC platforms like Coursera and Udemy rely on passive video watching, where students follow tutorials without hands-on workplace troubleshooting.
- **Static Job Simulations:** Platforms like Forage provide corporate business scenarios, but submissions are static files (PDFs/PPTs) with no live code execution or automated testing.
- **Unguided Cloud IDEs:** Cloud development environments like GitHub Codespaces offer raw computing machines, but lack curriculum, simulated company context, and guided task backlogs.
- **Binary Auto-Graders:** Existing assessment tools test code on simple pass/fail string matches rather than providing in-depth, line-by-line code reviews on pull requests.

Disadvantages

Key drawbacks and limitations found in existing educational platforms.

- **Clean-Slate Bias:** Students only practice writing small programs from scratch, leaving them unprepared for understanding and fixing large existing codebases with technical debt.
- **No Real Business Context:** Assignments lack realistic business problems, Jira-style task tickets, and Architecture Decision Records (ADRs) used by engineering teams in companies.
- **Limited Review & Feedback:** Existing tools provide only simple pass/fail test results without line-by-line feedback on code quality, security rules, and clean coding standards.
- **Unverifiable Resumes:** Self-reported skills on resumes are easy to exaggerate and do not give recruiters proof of practical ability or problem-solving capability.
- **Fragmented Learning Tools:** Learners must switch between videos, coding websites, and GitHub without a single connected workplace workflow.

Proposed System

The WORKUP platform workflow connecting simulated companies, tasks, sandboxes, and reviews.

SYSTEM DESCRIPTION

WORKUP acts as a virtual workplace platform that immerses learners in simulated engineering companies. It provides realistic task tickets, executes code in isolated cloud sandboxes, and delivers automated line-level reviews to help students build practical engineering competence.

PROCESS FLOW

1. **Career & Role Selection:** Learner chooses a career family and specialized role (e.g., Backend Engineer, Platform SRE).
2. **Simulated Company:** Enters a simulated enterprise (e.g., FinFlow Technologies) with authentic business context.
3. **Task Ticket & ADR:** Receives a Jira-style task ticket with acceptance criteria and Architecture Decision Records.
4. **Isolated Cloud Sandbox:** Opens a zero-setup browser workspace with path-jailed file execution.
5. **Implementation & Tests:** Refactors code and executes in-sandbox automated tests (Vitest) to verify correctness.
6. **AI Code Review:** Submits a pull request; the review engine evaluates code diffs and provides line-level feedback.
7. **Evidence of Skill:** Approved tasks generate an immutable portfolio backed by verified Git commit SHAs.

Advantages

Key advantages of the proposed WORKUP platform for learners and institutions.

- **Realistic Workplace Experience:** Learners work inside simulated companies (such as FinFlow Technologies) and solve genuine business problems.
- **Connected Project Workflows:** Follows the complete engineering cycle from task tickets to testing, pull request review, and portfolio evidence.
- **Zero-Setup Cloud Sandboxes:** Students write and test code directly in the browser without installing complex software on their personal computers.
- **Automated In-Sandbox Testing:** Built-in test execution (Vitest) allows learners to verify their code locally before submitting their work.
- **Line-by-Line Code Review:** An automated AI Tech Lead reviewer inspects code diffs and provides specific comments on code quality and architecture.
- **Verifiable Proof of Skills:** Replaces self-reported resume claims with an Evidence-of-Skill portfolio backed by verified Git commits and test logs.

TECNNOLOGY

Core technologies used in the development of the WORKUP platform.

TECNNOLOGY	USE
Next.js 14	Builds the full-stack web application with App Router, server-rendered pages, and secure API routes.
React.js	Creates the interactive user interface, including career catalogs, task boards, and workbench panels.
TypeScript	Provides static type-checking across frontend components, API handlers, and backend business logic.
Tailwind CSS	Provides responsive styling and modern layout components across all platform pages.
Prisma ORM	Manages database queries and models 21 relational domain entities (users, companies, tasks, workspaces, portfolios).
PostgreSQL / SQLite	Stores application data including user accounts, simulated company details, task tickets, and evaluation records.
Node.js Sandbox	Runs student code safely in path-jailed workspaces with strict execution timeouts (4000ms) to protect the server.
Vitest	Executes automated unit and integration tests inside the sandbox and checks student code against test assertions.
JWT (jose) & Bcryptjs	Handles user authentication with encrypted password hashing and secure HTTP-only session tokens.
Git Version Control	Captures code changes (diffs), generates commit hashes, and tracks student progress.

THANK YOU

Major Project Review — Stage-I | Questions & Discussion.

THANK YOU!

Special thanks to our Project Guide Mr. V. Veeresh, Head of Department Dr. M. V. Vijaya Saradhi, and the Department Review Committee.

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We welcome your valuable feedback and questions.